

- We have derived the cost functions. The cost function gives the minimum amount of cost incurred in production of a given amount of output. We get it from the cost minimization subject to a given level of output.

- In the example done in the last class , we get the following cost function when both labour and capital can be varied

$$c(w, r, y) = y^{\frac{1}{\beta+\alpha}} \Lambda,$$

$$\text{where } \Lambda = \left(w \cdot \left(\frac{r\alpha}{w\beta} \right)^{\left(\frac{\beta}{\beta+\alpha} \right)} + r \cdot \left(\frac{w\beta}{r\alpha} \right)^{\left(\frac{\alpha}{\beta+\alpha} \right)} \right)$$

- Suppose capital is fixed at some level $\bar{k}$. This is the minimum amount the producer has to install in the firm. In that case, the firm is always incurring a fixed cost of the amount $r\bar{k}$.

- In this case the cost function is

$$c(y) = w \cdot \left(\frac{y}{\bar{k}^\beta} \right)^{\frac{1}{\alpha}} + r\bar{k}.$$

$$c(y) = \gamma y^{\frac{1}{\alpha}} + F. \text{ Here } F \text{ is the fixed cost.}$$

- In this situation, the fixed cost component is always there since a fixed amount of capital is always needed. It cannot be varied in the short run.
- Suppose $k \leq \bar{k}$ and k can vary till $\bar{k}$. In this case in the same example as done in the class, $k = y^{\frac{1}{\beta+\alpha}} \left(\frac{w\beta}{r\alpha} \right)^{\left(\frac{\alpha}{\beta+\alpha} \right)}$ for $\bar{k} \geq y^{\frac{1}{\beta+\alpha}} \left(\frac{w\beta}{r\alpha} \right)^{\left(\frac{\alpha}{\beta+\alpha} \right)}$.
- We get a bound on y , it is $\bar{k}^{(\alpha+\beta)} \left(\frac{r\alpha}{w\beta} \right)^\alpha = y_{\bar{k}}$.
- The amount of capital employed is $k = \bar{k} = y^{\frac{1}{\beta+\alpha}} \left(\frac{w\beta}{r\alpha} \right)^{\left(\frac{\alpha}{\beta+\alpha} \right)}$ when the output $y \geq \bar{k}^{(\alpha+\beta)} \left(\frac{r\alpha}{w\beta} \right)^\alpha = y_{\bar{k}}$.

- The cost function is $c(y) = y^{\frac{1}{\beta+\alpha}} \Lambda$, for $y \leq y_{\bar{k}}$
 where $\Lambda = \left(w \cdot \left(\frac{r\alpha}{w\beta} \right)^{\left(\frac{\beta}{\beta+\alpha} \right)} + r \cdot \left(\frac{w\beta}{r\alpha} \right)^{\left(\frac{\alpha}{\beta+\alpha} \right)} \right)$ and
 $c(y) = w \cdot \left(\frac{y}{\bar{k}^\beta} \right)^{\frac{1}{\alpha}} + r\bar{k}$ for $y > y_{\bar{k}}$
- When the output is higher than $y_{\bar{k}}$ there is a component of fixed cost.

- Types of cost: variable cost, fixed cost, sunk cost.
- Variable cost: varies with the level of output. If the level of output increases, the variable cost always increases.
- Fixed cost. A fixed amount of cost independent of the level of output. Even if the level of output is zero, the producer bears fixed cost.
- Sunk cost: cost which cannot be recouped. Like amount of money spent on market research before entering a market. While refurbishing the firm: one example can be the amount spent on painting, but furnishers can be resold and some amount of expenditure can be recouped.

- Most common form of cost function is $c(y) = c_v(y) + F$. Here $c_v(y)$ is the variable cost component and F is the fixed cost.
- Average cost: cost per unit of output. $\frac{c(y)}{y} = \frac{c_v(y) + F}{y} = AC$
- $AC = \frac{C_v(y)}{y} + \frac{F}{y}$. Here $\frac{C_v(y)}{y}$ is average variable cost (AVC) and $\frac{F}{y}$ average fixed cost (AFC).
- Nature of average variable cost curves and average fixed cost curve.

- Marginal Cost (MC)= The additional cost incurred if output is increased by one more unit at the margin. $c(y) = c_v(y) + F$ and $c(y + 1) = c_v(y + 1) + F$.
 $MC = \Delta c(y) = c(y + 1) - c(y) = c_v(y + 1) - c_v(y)$.
 $\frac{dc(y)}{dy} = \frac{dc_v(y)}{dy}$.
- The nature of marginal cost curves

Relationship between average cost curve and marginal cost curve

$$\frac{d\left(\frac{c(y)}{y}\right)}{dy} = \frac{y \frac{dc(y)}{dy} - c(y)}{y^2} = \frac{MC - AC}{y}$$

When average cost curve is falling, the marginal cost curve must lie below the average cost curve. When average cost curve is rising, the marginal cost curve must lie above the average cost curve. So average cost curve must intersect marginal cost curve in the minimum point of the average cost curve.

The relationship is same between average variable cost curve and marginal cost curve.